Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

CompTIA Cybersecurity Analyst (CySA+) Training

TGS Ref No: TGS-2024049211

Version 2.1

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Document Version Control Record

Version	Effective Date	Summary of Changes	Author
1.0	6 July 2026	First edition — 5-day lesson plan aligned to the 30 hands-on CySA+ labs, with slide-number references to the Learner Guide Slides deck.	Tertiary Infotech Academy Pte Ltd
2.0	18 Aug 2026	Second edition — courseware updated to v7 Slides: full theory content from the legacy deck reordered to the four CS0-003 exam domains, exam domains shown up front, practice exam and CompTIA CySA+ exam registration added at the end.	Tertiary Infotech Academy Pte Ltd
2.1	18 Aug 2026	House-format fixes — daily schedule corrected to 9:30am-6:30pm with a 1-hour lunch (8 instructional hours/day), theory-delivery rows added per day, slide-reference summary corrected, Table of Contents and 11pt body applied.	Tertiary Infotech Academy Pte Ltd

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Course Details

Field	Detail
Course Title	CompTIA Cybersecurity Analyst (CySA+) Training
Course Code	TGS-2024049211
WSQ TSC	Cyber Risk Management (ICT-SNA-4007-1.1)
Duration	5 Days (40 hours)
Daily Schedule	9:30 AM – 6:30 PM (8 instructional hours/day)
Lunch Break	1:00 PM – 2:00 PM (1 hour)
Breaks	Short tea breaks are scheduled within each day's training hours
Delivery Mode	Instructor-led presentation, live demonstration and hands-on labs (Killercoda / free tools)
Assessment	Written Assessment (SAQ) 1 hr + Case Study (CS) 1 hr — Open Book (Day 5)
Prerequisites	Basic networking and Linux command-line familiarity; interest in cybersecurity operations
Slide Deck	CompTIA Cybersecurity Analyst (CySA+) Training-v8.pptx (919 slides)

Note: the 'Slides' column in each daily schedule references the exact slide numbers in the Learner Guide Slides deck for that activity.

Course Overview

This 5-day hands-on course develops practical cybersecurity analyst skills aligned to the CompTIA CySA+ CS0-003 exam. Across four domains — Security Operations, Vulnerability Management, Incident Response, and Reporting & Communication — learners complete 30 step-by-step labs using the free Killercoda playground and free industry tools, culminating in a written and case-study assessment.

Learning Outcomes

By the end of this course, participants will be able to:

- Identify cyber security risks, threats, and vulnerabilities, and assess their impact on the organization.
- Develop cyber risk assessments by consolidating insights from business to identify security gaps.
- Identify possible treatments for identified cyber risks, threats, and vulnerabilities.
- Implement endorsed treatment measures to address security gaps and ensure data protection.

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Day 1

Topic 1: Security Operations

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:50	Digital Attendance (AM); Trainer & Learner Introduction; Ground Rules	20 min	2–7
9:50 – 10:30	Lesson Plan; Skills Framework; Learning Outcomes; CS0-003 Exam Domains; Course Outline	40 min	9–19
10:30 – 10:45	Briefing for Assessment; Assessment; Criteria for Funding	15 min	20–23
10:45 – 11:00	*Tea Break*	*15 min*	—
	Topic 1: Security Operations — Topic Overview	**—**	**24–25**
11:00 – 1:00	Theory: Security Operations concepts	120 min	26–269
1:00 – 2:00	*Lunch Break*	*60 min*	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:05	Digital Attendance (PM)	5 min	—
	Hands-On Labs 1–10	**—**	**270–316**
2:05 – 4:00	Lab 1: Log Ingestion and Time Synchronization; Lab 2: OS Hardening and System Process Inspection; Lab 3: Network Segmentation and Zero Trust; Lab 4: Detecting Malicious Network Activity; Lab 5: Host-Based IOC Hunting	115 min	270–292
4:00 – 4:15	*Tea Break*	*15 min*	—
4:15 – 6:25	Lab 6: Packet Capture for Threat Hunting; Lab 7: SIEM Log Correlation; Lab 8: Email Header Analysis (SPF/DKIM/DMARC); Lab 9: Malware Triage with Hashing and VirusTotal; Lab 10: Threat Intelligence and MITRE ATT&CK	130 min	293–316
6:25 – 6:30	Day 1 Recap; Q&A	5 min	311–316
6:30	End of Day 1	—	—

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Day 2

Topic 2: Vulnerability Management

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:40	Digital Attendance (AM); Recap of previous day	10 min	—
	Topic 2: Vulnerability Management — Topic Overview	**—**	**317–318**
9:40 – 10:45	Theory: Vulnerability Management concepts (part 1)	65 min	319–452
10:45 – 11:00	*Tea Break*	*15 min*	—
11:00 – 1:00	Theory: Vulnerability Management concepts (part 2)	120 min	453–585
1:00 – 2:00	*Lunch Break*	*60 min*	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:05	Digital Attendance (PM)	5 min	—
	Hands-On Labs 11–19	**—**	**586–626**
2:05 – 4:00	Lab 11: Asset Discovery with Nmap; Lab 12: Vulnerability Scanning with OpenVAS (Greenbone CE); Lab 13: Web App Scanning with OWASP ZAP; Lab 14: Web Recon with Nikto and Burp Suite; Lab 15: Metasploit Framework Basics	115 min	586–608
4:00 – 4:15	*Tea Break*	*15 min*	—
4:15 – 6:25	Lab 16: CVSS Scoring and Prioritization; Lab 17: Exploiting and Mitigating XSS and SQL Injection; Lab 18: Patch Management and Hardening; Lab 19: Attack Surface Reconnaissance	130 min	609–626
6:25 – 6:30	Day 2 Recap; Q&A	5 min	622–626
6:30	End of Day 2	—	—

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Day 3

Topic 3: Incident Response and Management

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:40	Digital Attendance (AM); Recap of previous day	10 min	—
	Topic 3: Incident Response and Management — Topic Overview	**—**	**627–628**
9:40 – 10:45	Theory: Incident Response and Management concepts (part 1)	65 min	629–678
10:45 – 11:00	*Tea Break*	*15 min*	—
11:00 – 1:00	Theory: Incident Response and Management concepts (part 2)	120 min	679–727
1:00 – 2:00	*Lunch Break*	*60 min*	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:05	Digital Attendance (PM)	5 min	—
	Hands-On Labs 20–25	**—**	**728–757**
2:05 – 4:00	Lab 20: Cyber Kill Chain and ATT&CK Mapping; Lab 21: Evidence Acquisition and Chain of Custody; Lab 22: Memory Forensics with Volatility	115 min	728–742
4:00 – 4:15	*Tea Break*	*15 min*	—
4:15 – 6:25	Lab 23: Log Analysis for Incident Response; Lab 24: Containment with Host Isolation; Lab 25: Incident Response Playbook and Tabletop	130 min	743–757
6:25 – 6:30	Day 3 Recap; Q&A	5 min	752–757
6:30	End of Day 3	—	—

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Day 4

Topic 4: Reporting and Communication

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:40	Digital Attendance (AM); Recap of previous day	10 min	—
	Topic 4: Reporting and Communication — Topic Overview	**—**	**758–759**
9:40 – 10:45	Theory: Reporting and Communication concepts (part 1)	65 min	760–815
10:45 – 11:00	*Tea Break*	*15 min*	—
11:00 – 1:00	Theory: Reporting and Communication concepts (part 2)	120 min	816–870
1:00 – 2:00	*Lunch Break*	*60 min*	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:05	Digital Attendance (PM)	5 min	—
	Hands-On Labs 26–30	**—**	**871–889**
2:05 – 4:00	Lab 26: Vulnerability Management Report; Lab 27: Executive Incident Report; Lab 28: Security Metrics Dashboard (MTTD / MTTR)	115 min	871–881
4:00 – 4:15	*Tea Break*	*15 min*	—
4:15 – 6:25	Lab 29: Compliance Reporting (PCI DSS / ISO 27001 / NIST CSF); Lab 30: Stakeholder Communication and Lessons Learned	130 min	882–889
6:25 – 6:30	Day 4 Recap; Q&A	5 min	886–889
6:30	End of Day 4	—	—

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Day 5

Revision, Practice Exam and Final Assessment

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:40	Digital Attendance (AM)	10 min	—
9:40 – 10:45	Revision: Topic 1 Security Operations & Topic 2 Vulnerability Management	65 min	24–316, 317–626
10:45 – 11:00	*Tea Break*	*15 min*	—
11:00 – 1:00	Revision: Topic 3 Incident Response & Topic 4 Reporting	120 min	627–757, 758–889
1:00 – 2:00	*Lunch Break*	*60 min*	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:05	Digital Attendance (PM)	5 min	—
2:05 – 3:15	Practice Exam (warm-up quiz) and review	70 min	891–906
3:15 – 3:45	CompTIA CySA+ exam registration walkthrough (Pearson VUE)	30 min	907–910
3:45 – 4:00	*Tea Break*	*15 min*	—
4:00 – 4:15	Course Feedback & TRAQOM Survey; Assessment Briefing; Digital Attendance (Assessment)	15 min	911–919
4:15 – 6:15	Final Assessment: Written (SAQ) 1 hr + Case Study (CS) 1 hr	120 min	—
6:15 – 6:30	Assessment wrap-up; Recommended Courses; Support	15 min	—
6:30	End of Class	—	—

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Slide Reference Summary

Section	Content	Slides
Front Admin	Title, Attendance, Trainer, Ground Rules, Download Material, Lesson Plan, Skills Framework, Learning Outcomes, CS0-003 Exam Domains, Course Outline, Briefing, Assessment, Funding	1–23
Topic 1: Security Operations	Section + Overview + Theory + Labs 1–10 + Recap	24–317
Topic 2: Vulnerability Management	Section + Overview + Theory + Labs 11–19 + Recap	317–627
Topic 3: Incident Response and Management	Section + Overview + Theory + Labs 20–25 + Recap	627–758
Topic 4: Reporting and Communication	Section + Overview + Theory + Labs 26–30 + Recap	758–890
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Lab 2	OS Hardening and System Process Inspection	275–278
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Lab 10	Threat Intelligence and MITRE ATT&CK	311–316
Lab 11	Asset Discovery with Nmap	586–590
Lab 12	Vulnerability Scanning with OpenVAS (Greenbone CE)	591–595
Lab 13	Web App Scanning with OWASP ZAP	596–599
Lab 14	Web Recon with Nikto and Burp Suite	600–603
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Lab 19	Attack Surface Reconnaissance	622–626
Lab 20	Cyber Kill Chain and ATT&CK Mapping	728–732
Lab 21	Evidence Acquisition and Chain of Custody	733–737
Lab 22	Memory Forensics with Volatility	738–742
Lab 23	Log Analysis for Incident Response	743–747
Lab 24	Containment with Host Isolation	748–751
Lab 25	Incident Response Playbook and Tabletop	752–757
Lab 26	Vulnerability Management Report	871–874
Lab 27	Executive Incident Report	875–877
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	/ MTTR)	
Lab 29	Compliance Reporting (PCI DSS / ISO 27001 / NIST CSF)	882–885
Lab 30	Stakeholder Communication and Lessons Learned	886–889
Practice Exam	Practice section, warm-up quiz and answer key	891–906
Exam Registration	CompTIA CySA+ exam facts and Pearson VUE registration steps	907–910
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